

An explainable artificial intelligence model for predictive maintenance and spare parts optimization

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ABSTRACT

Maintenance strategies are vital for industrial and manufacturing systems. This study considers a proactive maintenance strategy and emphasizes using analytics and data science. We propose an Explainable Artificial Intelligence (XAI) methodology for predictive maintenance. The proposed method utilizes a machine learning project cycle and Python libraries to interpret the results using the Local Interpretable Model-agnostic Explanations (LIME) method. We also introduce an early concept of spare parts management, presenting insights from predictive maintenance outcomes and providing explanations for decision-makers to enhance their understanding of the influential factors behind predictions. This study demonstrates that utilizing machine learning models in predictive maintenance is highly beneficial; however, the binary outcomes of these models can be misunderstood by decision-makers. Detailed explanations provided to decision-makers will directly impact maintenance decisions and improve spare part management.

1. Introduction

With the evolution of technology and industrial progress, a substantial transformation has transpired in the concept of maintenance [1–3]. As the level of technology used increases, any defect or malfunction in machines has become more likely to lead to crucial problems [4]. Consequently, companies explore diverse methodologies in the pursuit of minimizing the losses caused by wrong strategies [5].

Maintenance is fundamentally the process of ensuring that all systems, machines, equipment, or vehicles in an organizational framework operate at maximum performance and make it sustainable. Any failure in the maintenance poses a significant risk, engendering critical financial losses, diminishing product quality, and escalating operational expenditures. Productivity and efficiency decrease and disruption even in the slightest part of the supply chain can have a negative impact on the entire chain.

The design and implementation of an effective maintenance process commence with determining an appropriate maintenance strategy. The maintenance strategy is determined by evaluating maintenance policies, equipment updates, resource allocation, provision of spare equipment, and decisions regarding inspection, repair, and replacement in a holistic

manner [6]. Various maintenance strategies exist. Initially, Bateman [7] classified them into three categories: Reactive, Preventive, and Predictive Maintenance. Then Weil [8] introduced a new category, Total Productive Maintenance, to Bateman's classification. Swanson [9], on the other hand, divided maintenance strategies into Reactive, Proactive, and Aggressive Maintenance. Tinga [6] refined these processes by sub-categorizing and established the widely accepted classification. Fig. 1 depicts a simplified version of Tinga's classification.

Reactive maintenance aims to conduct maintenance after any faults or defects occur and it is divided into two as Corrective and Detective Maintenance. Corrective Maintenance is maintenance performed when an equipment breakdown or it's observed that any failure or defect is going to happen [6,10]. On the other hand, proactive maintenance focuses on acting before any faults arise and it is forked into two as Preventive and Opportunistic Maintenance. Opportunistic maintenance is the strategy of including another equipment that needs maintenance in the maintenance process while maintaining any equipment [11]. Preventive maintenance is the maintenance approach that aims to replace or repair components before a malfunction occurs. It is generally used in systems where the consequences of damage may have critical consequences [6]. In aggressive maintenance, the entire system is

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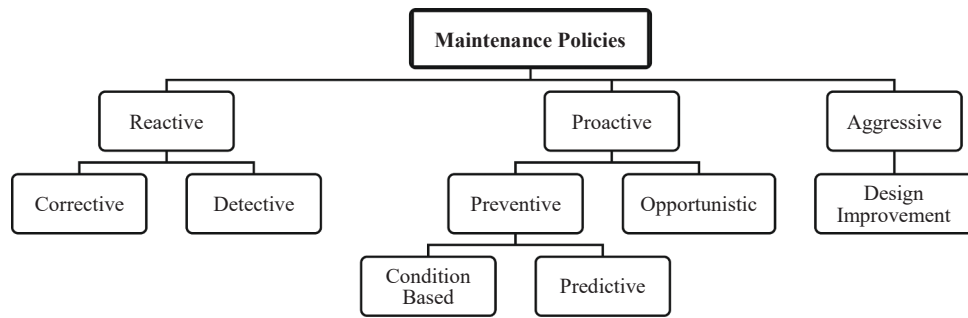


Fig. 1. Classification of Maintenance Strategies - Adapted from Tinga [6].

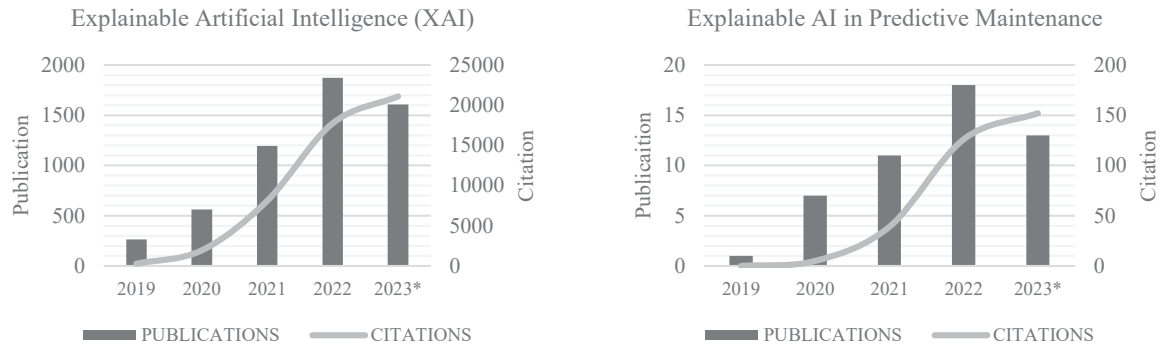


Fig. 2. Publications over time and number of times cited on Web of Science Database (Search parameters: “(((TS=(explainable ai)) OR TS=(xai)) OR TS=(explainable artificial intelligence)) AND TS=(predictive maintenance)”, Access Date: 28.11.2023).

reconsidered, redesigned, and improved to reduce the number of equipment failures. Tinga [6] linked Total Productive Maintenance as the subtype of Aggressive Maintenance. It is a maintenance policy where all employees in the system participate in the maintenance process, employees are responsible for the maintenance of the equipment they work with and aim to increase efficiency by preventing failures [12,13].

The challenge in the proactive approach lies in predicting faults in advance. Therefore, technologies such as monitoring systems, advanced statistical analysis, data analysis, machine learning models, and artificial intelligence are employed to predict potential malfunctions that have not occurred yet. Machine learning models can learn complex relationships and patterns by analyzing large datasets obtained from monitoring systems. Established models can be effective in determining the timing of faults. Modern technology has become a significant tool in making proactive maintenance strategies, which have been studied for many years, more effective and efficient.

Artificial Intelligence (AI) integration in our lives, particularly during the Industry 4.0 era, has brought raging changes. "Godfather of AI", Geoffrey Hinton explains AI's general functioning, the complexity becomes challenging to comprehend, much like the workings of the human brain. He associates designing the learning algorithm with establishing the principles of evolution. Once interacting with data, the algorithm generates complex neural networks, the precise mechanisms of which remain not fully understood [14]. Despite its advantages, the complexity of AI models, often treated as "black boxes," has raised concerns about transparency, leading to a noteworthy discussion on Explainable Artificial Intelligence (XAI) [15]. The term "black box" was originally introduced by Van Lent et al. [16] to characterize their system's capacity to elucidate the actions of AI-controlled entities within simulation games. It basically refers to artificial intelligence systems where users cannot see the operations. They often involve complex neural networks with thousands of artificial neurons, making it challenging for users to comprehend their decision-making processes.

The growing popularity of XAI is evident, as shown in Fig. 2, however, there is a noticeable lack of studies concerning the application of

XAI in Predictive Maintenance. Despite the widespread use of machine learning models in predictive maintenance, the importance of the explainability concept seems to be ignored in this domain. This study aims to discuss the explainability of a Ground Support Equipment (GSE) predictive maintenance application for a Maintenance, Repair, and Overhaul (MRO) company, while simultaneously establishing a framework for Spare Parts Management.

The research process involves a literature review to understand existing XAI and predictive maintenance studies. Then simple data analysis lifecycle approach and various machine learning algorithms are introduced in the methodology section. Then an XAI technique, LIME was executed to enhance model explainability to begin the creation of a foundational framework for spare parts management based on predictive maintenance outcomes. This effort aims to contribute valuable insights to the field and promote the next phase in the evolution of spare parts management.

2. Literature Survey

As AI becomes more prevalent in critical decision-making domains, the shift towards inherently interpretable models gains importance to address risks and maintain societal trust [17,18]. To tackle challenges posed by non-transparent models, recent studies underscore the necessity for clarity in machine learning models [19]. Although progress in XAI is apparent, the field is considered immature, lacking formal definitions, and encountering integration challenges into machine learning workflows [19]. Notable practical applications, such as the use of XAI in predictive maintenance in manufacturing [20], literature review on XAI in the domain of digital twin applications [21], and the development of novel tools for interpretability in tree-based models [22], showcase the evolving concepts within XAI. In the healthcare domain, interpretability remains crucial, as emphasized in a survey exploring AI and machine learning advancements in 2021 [23]. Moreover, in supply chain management, different risk categories can be studied using artificial intelligence and machine learning algorithms [24].

The central role of AI and XAI has become increasingly important, particularly in autonomous task execution and providing human-understandable explanations in the industrial landscape. This situation does not only anticipate their impact across industries but also recognizes challenges and outlines future research directions [25].

The emergent field of explainable artificial intelligence is pivotal in enhancing transparency and understanding in AI systems. As AI technologies become increasingly integrated into various domains, understanding the underlying mechanisms and maintenance strategies of XAI becomes essential to ensure reliability and sustained performance over the long term.

2.1. XAI in Predictive Maintenance

In this section, a review is conducted to explore the role and necessity of XAI, emphasizing its significance across various domains, with a particular focus on predictive maintenance. Examined studies are categorized into three groups.

The first group focuses on the challenges of implementing XAI in industrial applications, proposing solutions, and highlighting the importance of transparency, interpretability, and explainability. These studies provide insightful information by summarizing literature and offering fundamental knowledge for newcomers in the field [26–29]. Additionally, other recent studies address the application of artificial intelligence in organizations [30], in the context of the predictive maintenance domain [31], and Industry 4.0 [32] through systematic reviews.

The second group covers case studies where predictive maintenance and XAI are integrated into different domains. The studies contain valuable information regarding the application of machine learning models and XAI techniques. For instance, Mey and Neufeld [33] utilized the integration of CNN (Convolutional Neural Network) and three different XAI techniques (GradCAM (Gradient-weighted Class Activation Mapping), LRP (Layer-wise Relevance Propagation), and LIME) for vibration-based imbalance classification in an electric motor. Kim et al. [34] applied the isolation forest machine learning algorithm to detect anomalies for conditional monitoring of marine engines, and they explained the results using the SHAP (Shapley Additive Explanations) XAI technique. Baptista et al. [35] also presented an application of SHAP technique to enhance the explainability of data-driven prognostic approaches. Hrnjica and Softic [20] explored the combined use of gradient boosting machine learning model and XAI in a production domain dataset. Li et al. [36] discussed the importance of a Condition Based Maintenance strategy for fault diagnosis in aircraft fuel systems, examining XAI methods. Similarly, Gawde et al. [37] presented an explainable predictive maintenance strategy for multi-fault diagnosis of rotating machines using multi-sensor data fusion. Additionally, Ferraro [38], Sanakkayala [39], and Taşçı et al. [40] addressed the Remaining Useful Lifetime (RUL) estimation and XAI techniques in use to predict the remaining lifespan of HDDs, estimate bearing fault prognosis and as predictive maintenance application, respectively. Amram et al. [41] investigated the use of interpretable machine learning approaches for data-driven predictive maintenance, aiming to predict hard disk failures in data centers. Galanti et al. [42] propose a decision support system and Arena et al. [43] propose conceptual framework for predictive maintenance machine learning applications. Some other research has been done to forecast bike rental demand in urban areas [44], the medical insurance cost [45], and total revenue for investment decisions in the financial management domain [46].

The third group explores the human factor in the context of machine learning, predictive maintenance, and XAI concepts. Wellsandt et al. [47] and Ahmed et al. [25] emphasized a human-centric approach, examining hybrid augmented intelligence and responsible AI concepts. Turner et al. [48] established a connection between XAI and humans in the loop, highlighting the vital role of human empowerment in maintenance automation and presenting a framework. Langer et al. [49]

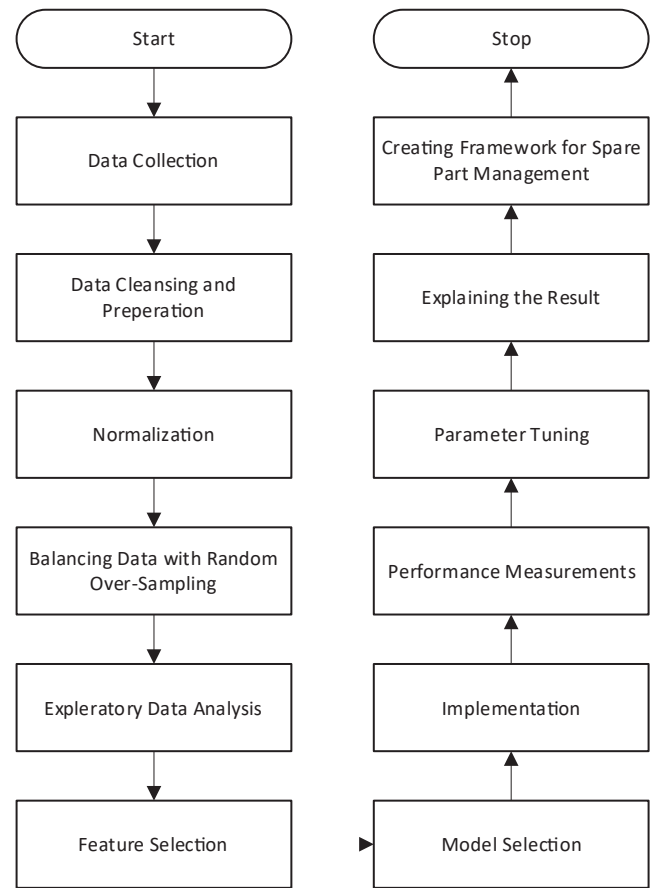


Fig. 3. Methodology.

examined the main stakeholder groups demanding explainability in artificial systems and their requirements, proposing a model to meet these demands. In the final study within this category, Gentile et al. [50] investigated the impact of explanations provided to humans on decision-making. In addition to traditionally employed normative and contrastive explanations, the study also examines a combined type, namely normative plus contrastive explanations. The experimental results demonstrate that the type of explanation has an influence on decision time, workload, hit rate, and sensitivity in human decision-making.

The reviewed studies indicate that, although the number of studies in the literature is limited, XAI can play a significant role in predictive maintenance. This paper has been approached with a holistic perspective. Considering the challenges described in the first group of literature, it suggests a simple yet effective solution. Given that transforming decision-makers from machines to humans is a primary focus of this study, it shows similarity with the third group of literature. Furthermore, it aims to contribute to the existing literature by exploring a new domain, Predictive Maintenance for GSE, within the second group of literature.

3. Methodology

Initially data collection process is undertaken and followed by the process of cleaning the irrelevant, duplicated, or outlier values from the data. Subsequently, normalization and balancing are employed to standardize variables in a common scale and ensure the data is balanced. When the data is fully ready for analysis, exploratory data analysis is conducted to learn any patterns and trends, and the feature selection process is applied. Model selection ensures the best-fitted model for the problem and is followed by performance evaluation and parameter

tuning to optimize model efficiency. Finally, and the most crucial part of the methodology, results are interpreted, and it gives insights from mathematical values to feed the decision-making process. The methodology employed in the study is illustrated in Fig. 3.

3.1. Problem definition

Machine learning models are utilized to predict whether equipment is faulty or not based on collected sensor data. Common applications generally provide results within the 0–1 range, and it becomes less effective as it approaches center. Machine learning models supported with XAI techniques provide the decision maker with more detailed, interpreted results. Despite various studies on XAI, its application in predictive maintenance for the GSE domain seems to be a novel target.

3.2. Data collecting

In this study, The Predictive Maintenance Dataset (AI4I) is used. This dataset which is synthetically generated by Matzka [51] and consists of 14 features with 10.000 rows of data, has been frequently studied in machine learning applications.

In real-life applications, condition monitoring systems such as systems, sensors, and services should be deployed to apply this methodology. Predictive maintenance typically involves parameters such as temperature, vibration measurements, ultrasonic testing, infrared thermography measurements, and oil analysis for effective implementation. In this study and dataset, multiple temperature, speed, and torque measurements are used to deploy the model.

3.3. Data cleansing and preparation

At this stage of the study, the dataset is cleaned from irrelevant observations and duplicates. Unwanted formatted data in the dataset should be updated. Missing data should be handled through appropriate techniques. Moreover, an analysis of outlier treatment is conducted to determine whether to remove outliers from the dataset, and actions are taken accordingly.

3.4. Normalization

Normalization is the process of setting a common scale for data that has different ranges and measurement metrics. It enables comparability among features. Min-max scaler has been run for normalization in this study with the following formula: $X_{\text{new}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$.

3.5. Balancing Data

Classification datasets generally tend to be unbalanced. Using such data in its raw form for the train model may lead to a biased result toward a class of features. In order to handle this situation and enhance accuracy various techniques such as under-sampling, over-sampling, and SMOTE (Synthetic Minority Over-sampling Technique) are employed in literature [52]. Over-sampling basically increases the size of the whole data without adding any variety, while SMOTE increases both the sample size and the variety. In this study, random over-sampling has been utilized since we do not need any variety in our data. Data points were randomly selected from the dataset and duplicated to balance the dataset in terms of 0 and 1.

3.6. Exploratory data analysis

In this stage, the dataset goes for a more detailed examination. The relationships between variables and their connection to the model are reviewed. Trends and patterns are investigated. The appropriate statistical method is chosen, and the results are analyzed and visualized [46,

		Actual Values	
		True	False
Predicted Values	True	True Positive (TP)	False Positive (FP)
	False	False Negative (FN)	True Negative (TN)

Fig. 4. Confusion Matrix.

53]. Univariate Analysis consists of statistical measurements such as central tendency, range, variance, and standard deviation. Histograms and bar charts are the visualization tools for univariate analysis. While Bivariate Analysis uses various statistical methods and visualization tools to explore relationships between variables, Multivariate Analysis on the other hand, involves more complex statistical methods and visualization tools to examine multiple variables together and understand all relationships in the model.

3.7. Feature selection

Feature Selection is a crucial component of feature engineering. When building a model, there are often numerous input variables. It is the process of analysis of which variables should be used as input to the model according to its impact on model performance. In this process, redundant or irrelevant features are eliminated. It leads to a reduction in overfitting problems and improves accuracy. Various techniques such as forward selection, backward elimination, and recursive feature elimination generally used.

3.8. Model selection and implementation

PyCaret is an open-source machine learning library in Python designed to automate the machine learning workflow [54]. With providing the necessary inputs and executing the library, multiple machine learning algorithms can be deployed in a single run and ranked based on predefined performance criteria with PyCaret. It can also be employed as a precursor in the model selection process and identify the best-performed models for special separation execution.

3.9. Performance Measurements

It is recognized that relying on the same aggregated measurements at all times is inadequate and cannot accurately represent the performance

Table 1
Performance Evaluation Metrics.

	Metric	Formula
Regression Metrics	MSE	$\frac{1}{n} \sum_{i=1}^n (Actual_i - Predicted_i)^2$
	RMSE	$\sqrt{\frac{1}{n} \sum_{i=1}^n (Actual_i - Predicted_i)^2}$
	MAE	$\frac{1}{n} \sum_{i=1}^n Actual_i - Predicted_i $
Classification Metrics	Accuracy	$\frac{(TP + TN)}{(TP + FP + FN + TN)}$
	Precision	$\frac{TP}{(TP + FP)}$
	Recall	$\frac{TP}{(TP + FN)}$
	F1 Score	$\frac{2TP}{(2TP + FP + FN)}$

Table 2
Dataset Description, Matzka [51].

Data	Description
UID	Unique identifier ranges from 1 to 10000
Type	Quality variants consist of a Low, Medium, or High product
Air Temperature [K]	Created through a random walk process and then standardized to have a standard deviation of 2 K centered around 300 K.
Process Temperature [K]	Created through a random walk process and then standardized to have a standard deviation of 1 K and increased the air temperature plus 10 K.
Rotational Speed [RPM]	Derived from a power of 2860 W and combined with normally distributed noise.
Torque [Nm]	Normal distribution around 40Nm with a standard deviation of 10 Nm and with no negative values.
Tool Wear [min]	Quality levels, H/M/L, contribute 5/3/2 minutes of tool wear to the utilized tool during the process.
Machine Failure	Label that shows whether the machine has breakdown or not.

of algorithms in the last decades [55]. In the comparison of model performances today, regression models typically use metrics such as MSE (Mean Squared Error), RMSE (The Root Mean Squared Error), MAE (Mean Absolute Error), and R^2 , while classification models utilize metrics like Accuracy, Precision, Recall, and F1 Score.

As can be expected, selecting the model with the best performance across all metrics is a desired target, but in real-world problems, trade-offs between different metrics may be necessitated.

Classification results are generally summarized by a confusion matrix, shown in Fig. 4, with crosstab labels as predicted and actual results. True positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) are the elements of this matrix. They represent the number of data points that are actually true but predicted as false predictions, or actually true, and also predicted as true, or vice versa [56]. Then classification metrics use these values as pivot elements as shown in Table 1.

In scenarios where prediction errors are equally important, Accuracy is used. It measures the accuracy of correct predictions across the total number of predictions. Precision represents the accuracy of positive predictions and aims to minimize false positives, while Recall focuses on correctly predicting positives and aims to minimize false negatives. The

F1 Score combines Recall and Precision and provides a single metric through the harmonic mean of them. Looking from a predictive maintenance perspective, the objective is to achieve a higher Recall value to accurately identify faults when detecting machine issues. Conversely, a higher Precision is desired to accurately identify non-failure situations. To balance the risk, considering both precision and recall and evaluating them together using the F1 Score is more appropriate. In this study, the F1 Score is used to compare models in order to capture all failure situations and avoid unnecessary maintenance activities.

3.10. Parameter tuning

Essentially, hyperparameter tuning is the process of selecting optimal values for a machine learning model's parameter. These parameters play a crucial role in fine-tuning the performance of the model, contributing to increased accuracy, generalization, and other relevant metrics. Various approaches, including manual tuning, random tuning, and grid tuning, are employed for this purpose in machine learning literature [57]. In this study, a cross-validation technique was performed with PyCaret to find the best hyperparameters for a model with a selected optimized metric as F1-score.

3.11. Explaining the result

Interpretability is a fundamental aspect of this paper, and it builds upon the introduction of Explainable AI in previous chapters. In the literature, commonly used Explainable AI (XAI) methods include SHAP, DeepSHAP, DeepLIFT (Deep Learning Important Features), CXplain, and LIME. In this study, the LIME method proposed by Ribeiro et al. [58] has been chosen due to its model-agnostic nature, user-friendly characteristics, and its ability to provide local explanations.

3.12. Creating the Framework for Spare Part Management

After the prediction of the machine learning model, an interface will be provided to the decision-maker, displaying the results along with explanations of the features and their values that influence the prediction. This way, the decision-maker can make maintenance decisions for

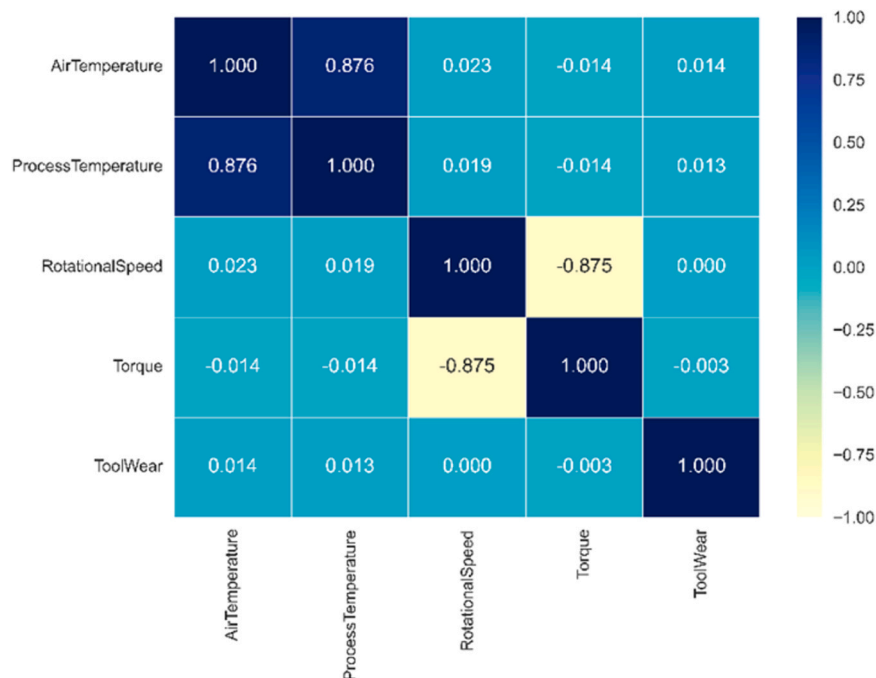


Fig. 5. Correlation Between Numeric Variables.

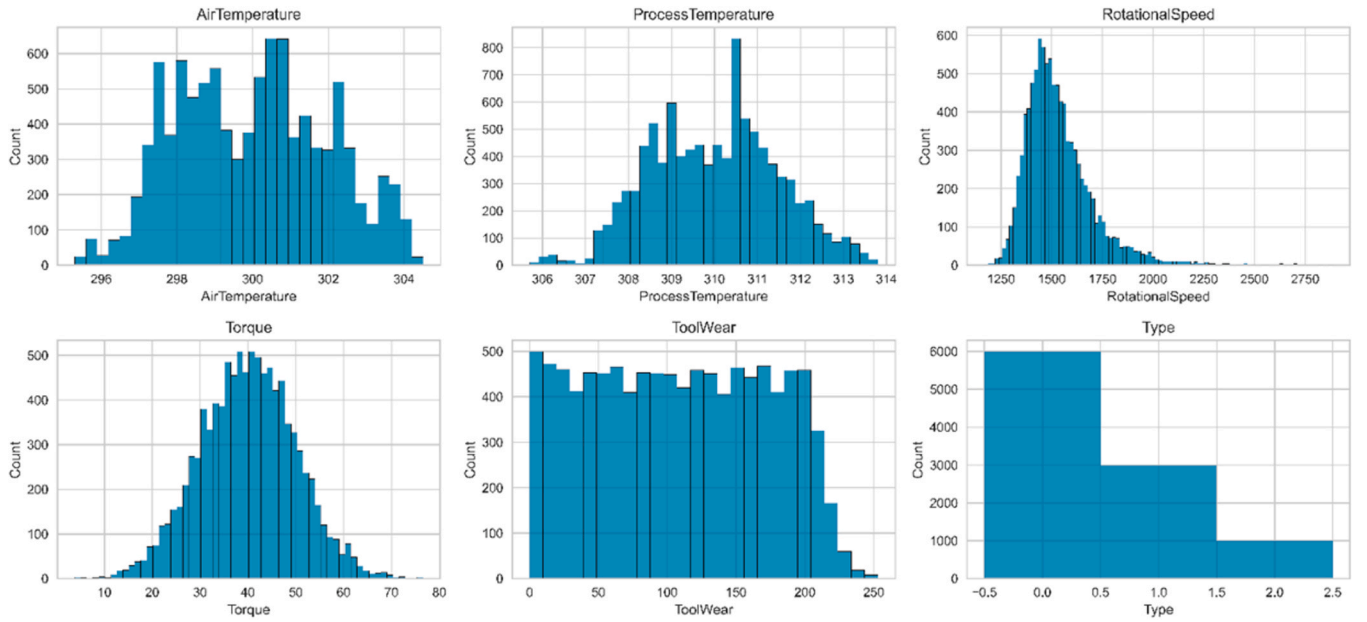


Fig. 6. Histograms of Numeric Variables.

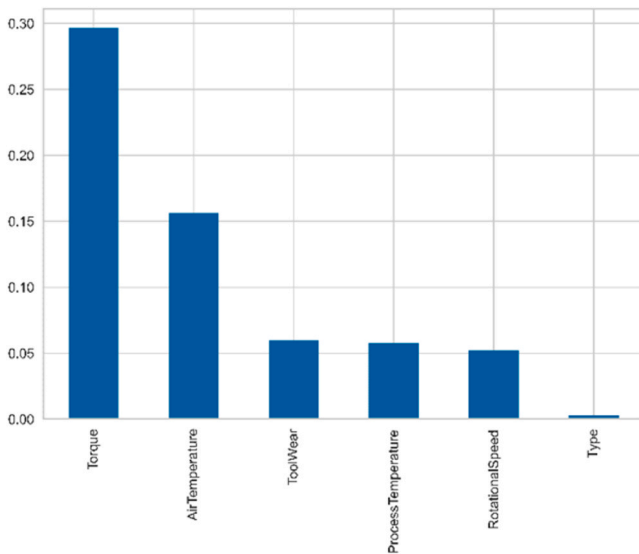


Fig. 7. Permutation Importance of Features.

equipment not only by looking at a single value in the 0–1 range but also by examining explanations that show the scores of important features.

4. Results

The proposed methodology has been implemented and Table 2 displays selected eight data headers and their descriptions from among the fourteen features included in the AI4I dataset, which has been frequently studied in machine learning applications. The remaining six features were excluded from the list due to being either irrelevant or duplicated headers.

It is observed that only 3.39 % of the entire data is labeled as a failure. This imbalance in the dataset has been addressed by performing random over-sampling, resulting in a balanced dataset.

As illustrated in Fig. 5, Air Temperature and Process Temperature variables and Rotational Speed and Torque variables display high correlation values. The relationship between Air Temperature and Process

Table 3
Top Features PPS.

Feature	Score
ToolWear	7173.12
Torque	909.62
RotationalSpeed	407.81
Type	12.14
AirTemperature	0.90
ProcessTemperature	0.09

Temperature variables could be a result of the synthetic nature of the dataset. This relationship should be verified by checking a real dataset, and the model performance can be re-evaluated accordingly. Additionally, an examination of the relationship between the Rotational Speed and Torque variables could provide insights, and an explanation can be derived. Alternatively, creating a new variable derived from these two variables is also a viable option.

In Fig. 6, histograms of numerical variables are displayed, while Fig. 7 illustrates the permutation importance of features.

As seen in Table 3, Tool Wear, Torque, and Rotational Speed features appear to have the most significant impact on the model. Although it was possible to proceed with these three features at this stage, no feature elimination was performed in the study, and all features were utilized because of their permutation importance values and limited number of features.

In Table 4, the results of running PyCaret are presented. The Extra Trees Classifier has yielded the best results across all metrics for the train data; however, after parameter tuning, the Ada Boost Classifier has the best-performed model based on the F1 score as shown in Table 5.

Although Ada Boost Classifier has ranked first place for the train data, its F1 score has decreased to around 82 % for the test data. This indicates potential overfitting. Thus, the other models are executed for the train data.

The second-ranked algorithm in Table 4, the Random Forest algorithm, was executed on the test data, and the results presented in Table 6 were obtained. According to these results, the model has outperformed the Ada Boost Classifier in terms of F1 Score for the test data (Weighted Average 0.98).

After determining a model with a satisfied performance level, efforts were directed toward the interpretability of the results. Predictions were

Table 4

PyCaret Comparison of Model Results.

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
et	Extra Trees Classifier	0.9985	1.0000	1.0000	0.9971	0.9985	0.9970	0.9970	0.0820
rf	Random Forest Classifier	0.9945	1.0000	1.0000	0.9892	0.9946	0.9891	0.9892	0.1060
lightgbm	Light Gradient Boosting Machine	0.9915	0.9993	1.0000	0.9833	0.9916	0.9830	0.9831	0.1080
dt	Decision Tree Classifier	0.9909	0.9909	1.0000	0.9822	0.9910	0.9819	0.9821	0.0120
knn	K Neighbors Classifier	0.9732	0.9895	1.0000	0.9491	0.9739	0.9464	0.9478	0.7100
gbc	Gradient Boosting Classifier	0.9628	0.9913	0.9891	0.9398	0.9638	0.9257	0.9270	0.1600
ada	Ada Boost Classifier	0.9299	0.9759	0.9373	0.9235	0.9304	0.8597	0.8598	0.0680
qda	Quadratic Discriminant Analysis	0.8468	0.9312	0.8028	0.8803	0.8397	0.6935	0.6963	0.0100
nb	Naive Bayes	0.8336	0.9036	0.8187	0.8441	0.8311	0.6673	0.6676	0.6280
lda	Linear Discriminant Analysis	0.8182	0.9045	0.8124	0.8221	0.8172	0.6364	0.6365	0.0100
ridge	Ridge Classifier	0.8167	0.0000	0.8105	0.8209	0.8156	0.6335	0.6336	0.0100
svm	SVM - Linear Kernel	0.8173	0.0000	0.7974	0.8311	0.8133	0.6346	0.6359	0.0120
lr	Logistic Regression	0.8133	0.9043	0.8104	0.8153	0.8128	0.6266	0.6267	0.7960
dummy	Dummy Classifier	0.5000	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0060

Table 5

Best Model After Parameter Tuning: Ada Boost Classifier.

Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
Ada Boost Classifier	0.9991	1.0000	1.0000	0.9983	0.9991	0.9983	0.9983

Table 6

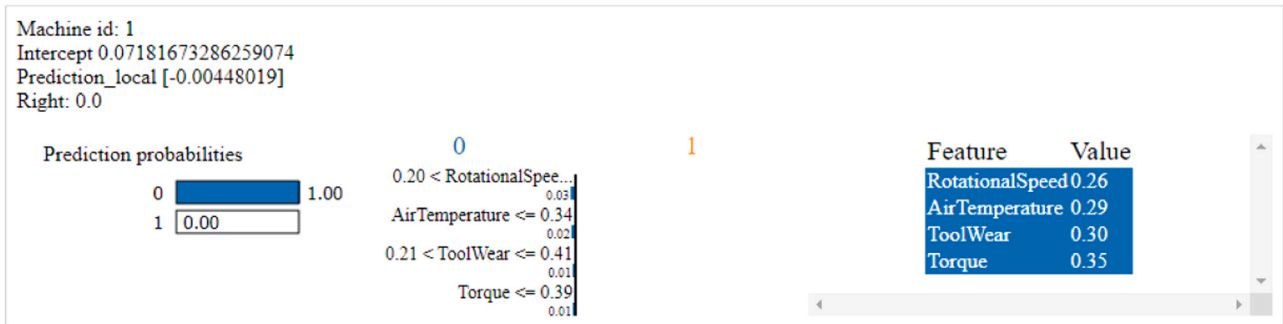
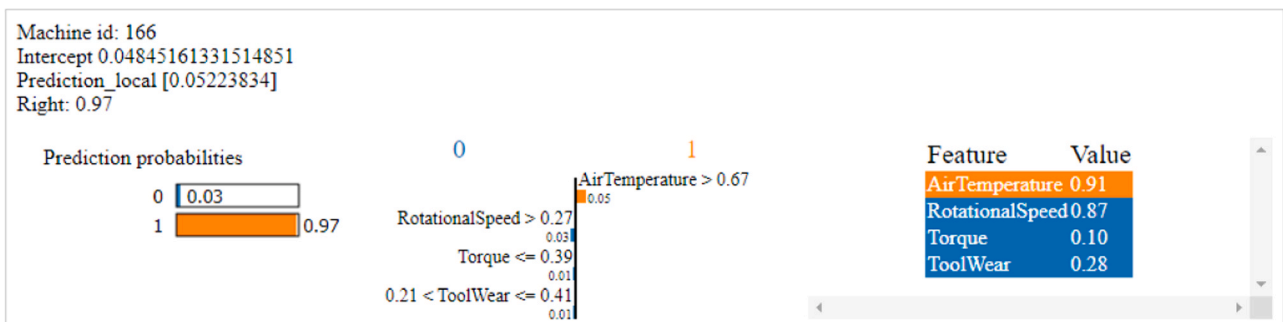
Classification Report of Random Forest Model.

	Precision	Recall	F1-score	Support
0 (non-failure)	0.99	0.99	0.99	1932
1 (failure)	0.76	0.62	0.68	68
Accuracy			0.98	2000
Macro Avg.	0.88	0.81	0.84	2000
Weighted Avg.	0.98	0.98	0.98	2000

explained using the LIME Python library's Tabular Explainer method. Figs. 8–11 show examples from these explanations. With these insights, the decision-maker will be better equipped to make maintenance decisions and initiate the spare parts procurement process.

Fig. 8 shows a machine that is predicted to be non-failure, and Fig. 9 shows a machine that is classified as a failure. However, as understood from the probabilities, these predictions are obvious. Maybe there is no reason for asking explanations for them but what can be said about the machines presented in Fig. 10 and Fig. 11? Can we assert with high confidence that the machine depicted in Fig. 10 is failure-free, or there is no way to delay the maintenance of the machine shown in Fig. 11?

Traditional machine learning applications, as they tend to round

**Fig. 8.** Explainer for Machine 1.**Fig. 9.** Explainer for Machine 166.

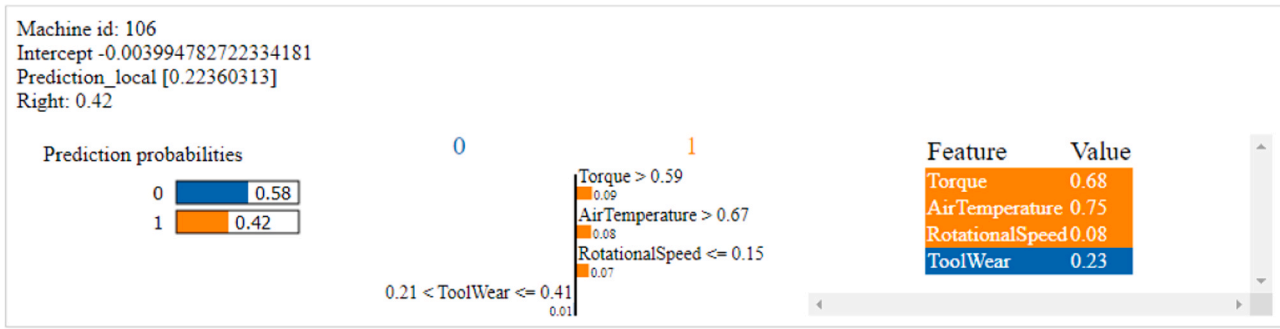


Fig. 10. Explainer for Machine 106.

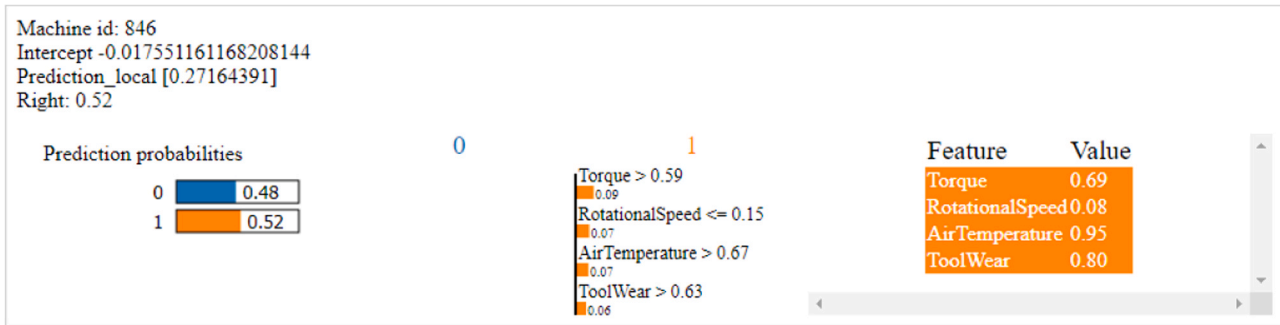


Fig. 11. Explainer for Machine 846.

these probabilities to either 0 or 1, do not offer explanations to the decision-maker. However, with Explainable AI, the decision-maker can examine Machine 106, enabling them to make timely maintenance decisions or prefer early procurement of spare parts for future maintenance, based on a more detailed understanding.

5. Conclusions

This study involves a comprehensive literature review to present insights into existing Explainable Artificial Intelligence (XAI) and predictive maintenance studies. The research process also includes the application of machine learning and XAI techniques, aiming to enhance model explainability and establish a foundational framework for Spare Parts Management.

The methodology of the study follows a basic data analysis and machine learning project life cycle and utilizes XAI tools to generate explanations for predictions. The PyCaret library is employed to explore models, and a random forest model that has an impressive 98 % F1 score on the test data, is built. Following the generation of model predictions, the LIME (Local Interpretable Model-Agnostic Explanations) method is applied to provide detailed explanations for decision-makers.

Despite the widespread use of the machine learning model, this study emphasizes the ignored importance of the explainability concept in the predictive maintenance domain. The speculative philosophy behind binary decisions (0 or 1) is clarified, and the necessity of providing explanations for predictions is emphasized with practical examples. Therefore, this study aims to be a joint node between Explainable AI, Predictive Maintenance, and Spare Part Management.

A simple limitation of the study is the utilization of a synthetic dataset (AI4I), even if it has been used many times in the literature. Running the model with the same methodology with the real-world data can easily handle this limitation.

Future research directions could be integrating this study with real-time monitoring systems and automating spare parts procurement processes based on real-time predictions and alert systems. This potential study holds promise for advancing the practical applications of

Explainable AI in the context of predictive maintenance and spare parts management.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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